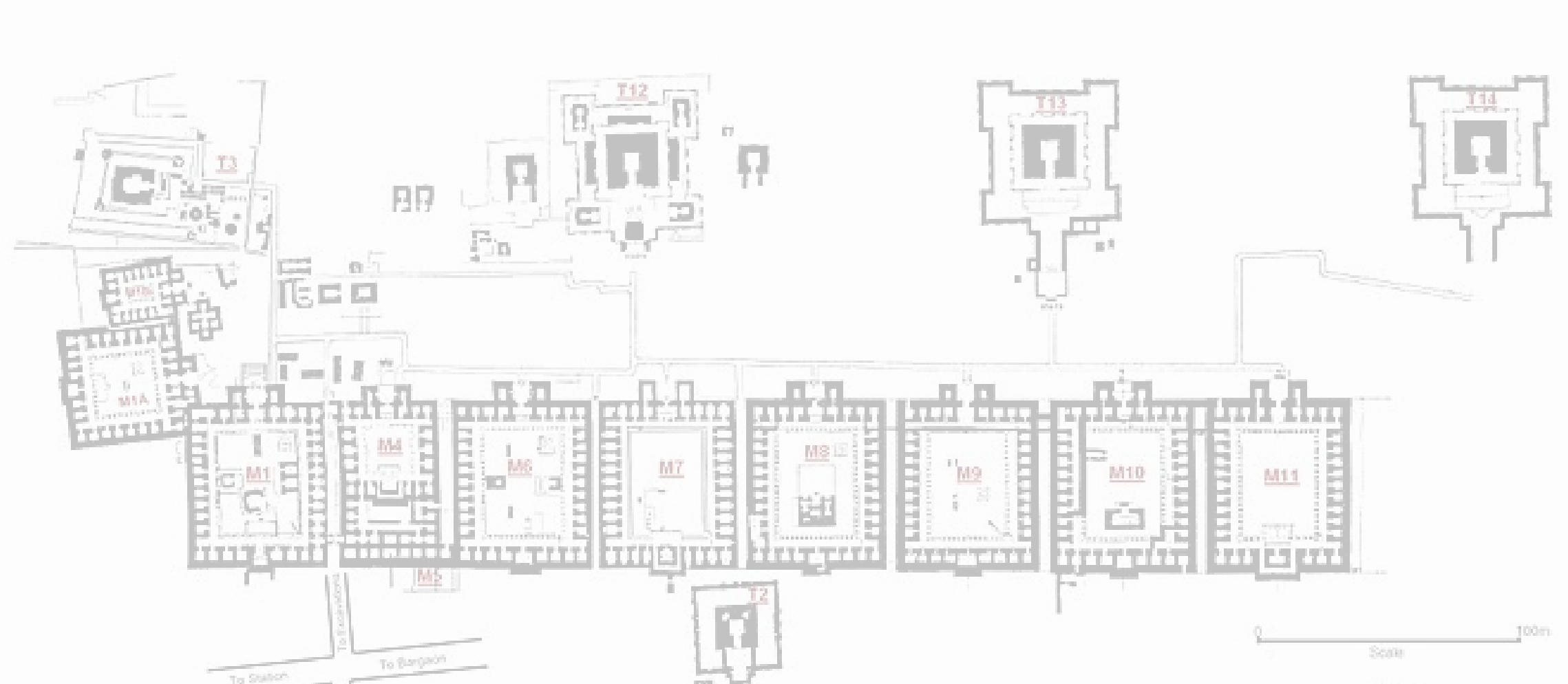




Nālandā
UNIVERSITY

Kshitish kumar(10), Prakhar Jain(35), Prateek Roongta(36)

Introduction

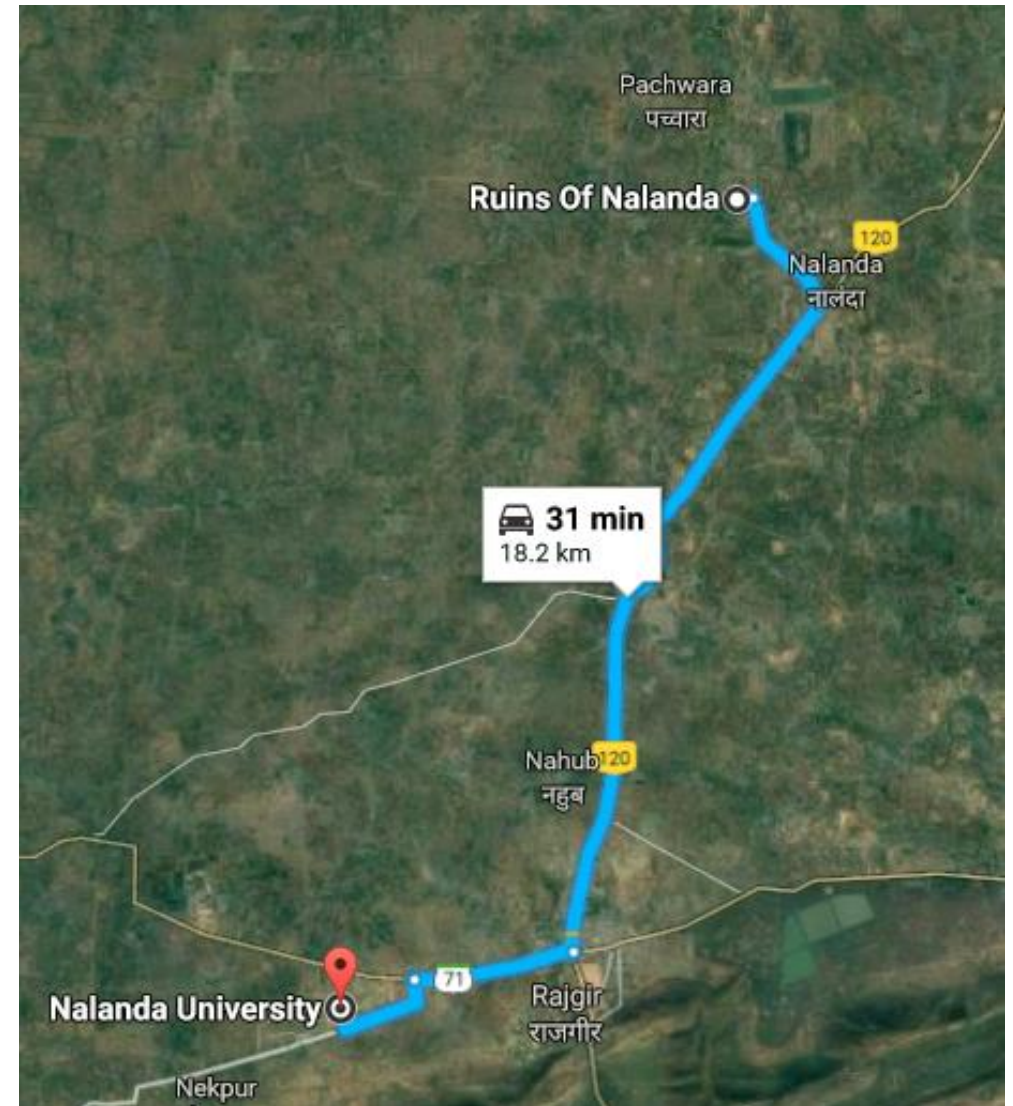
- Eight hundred years after the destruction of Nalanda, former President of India ***Dr. A.P.J. Abdul Kalam***, while addressing the Bihar State Legislative Assembly, in March 2006 mooted the idea of reviving the university.
- The State Government of Bihar adopted the visionary idea and consulted the Government of India on the way ahead. It also began its search for a suitable location for the new Nalanda University. It identified and acquired ***450 acres*** of land for the University in Rajgir, Bihar.
- Since the hallmark of the ancient Nalanda was its ***internationalism***, so the Government of India shared this proposal with the leaders of the East Asia Summit ('EAS') and got positive response for the establishment of the Nalanda University.

Type of the project – *Institutional project*

- Nalanda is an Institutional project which offers Master's degrees in three schools presently-
 1. The School of Historical Studies
 2. The School of Ecology and Environment Studies
 3. The School of Buddhist Studies, Philosophy and Comparative Religions.
- It is an educational space not only for Indians but it also offers education to the students of other countries like Japan, Iran, Greece Thailand etc.
- It is designed to accommodate 2500 students and 500 faculty members in the campus.

Location and Connectivity

- The University is located in **Rajgir**, a small hill town in Nalanda district of the state of Bihar in India. Scenic and green, Rajgir was the capital of ancient Magadh and was at that time called Rajgriha (the home of Royalty).
- Site can be access by NH-7 by road.
- A railway station and a airport is also proposed near the site which will make the site more approachable to the users.



New and old campus

Context

Geographical Context

- Entire land near the site has a backdrop of beautiful Rajgir hills.
- Site is surrounded by farm land and small villages of farmer.
- Hill are covered with dense tree which is a shelter for animals also.
- The design has to respect the natural beauty of the existing surrounding.
- Site is located on the foothill of Rajgir Hill. Hence design also has to consider the flow of water which will come from mountain.



Rajgir Hill

Historical Context

- Old Nalanda university is around 18 km away from the new proposed site.
- The new campus also has to give respect its origin which is very near to the site. There should be some similarity in new campus so that the same ambiance can be achieved.
- The entire land of that part has its spiritual significance for Buddhism, Jainism and Hinduism. A lot of holy places of these religion are located near the site.
- These three religion were one of the main subject in the old university. The new campus should give a reflection of spirituality also.



Ruins of Nalanda



Viswa santi stupa

Need of the project

- It is an ambitious project for Bihar and Indian Govt.
- President APJ Abdul Kalam while addressing the joint session of Bihar Vidhan Sabha, said that ***the glory of past must be retained and through re-establishing the University, India will get a chance to become global hub for education.***
- Also, other Asian countries were in support of Nalanda project.



Ruins of Nalanda

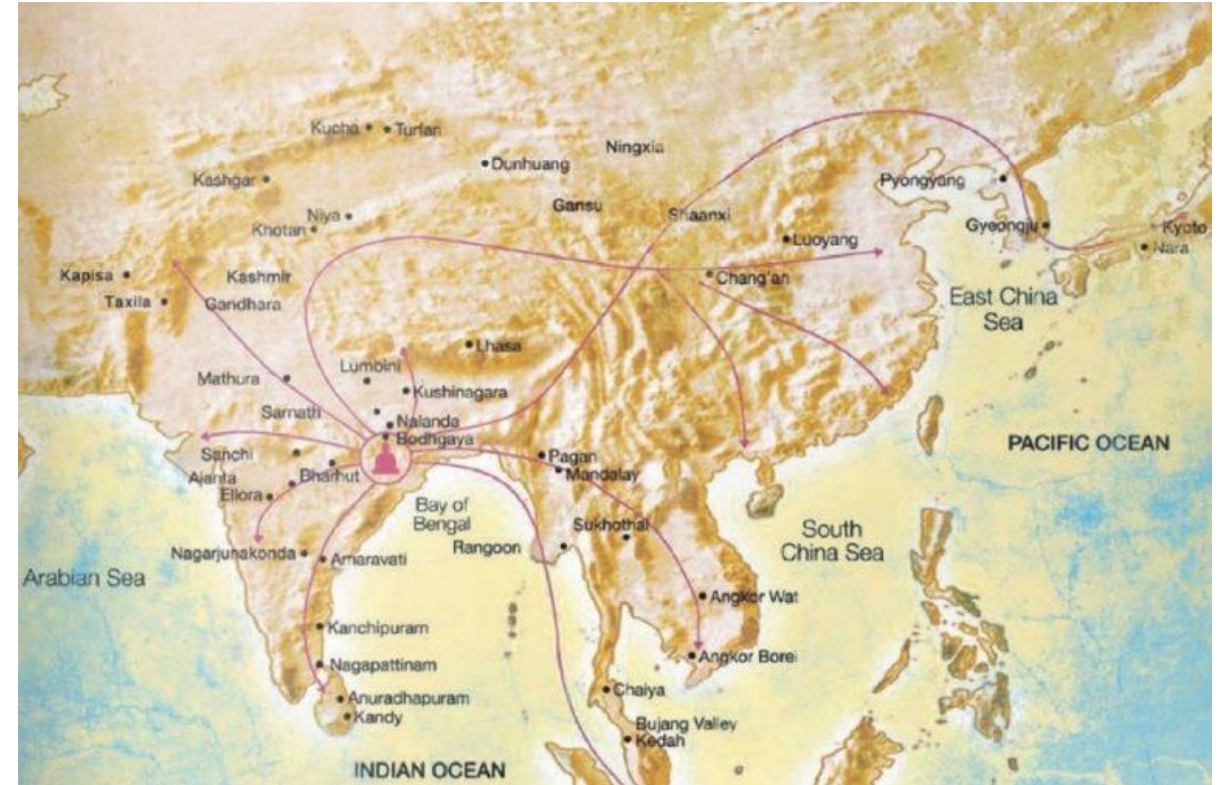
History of the project

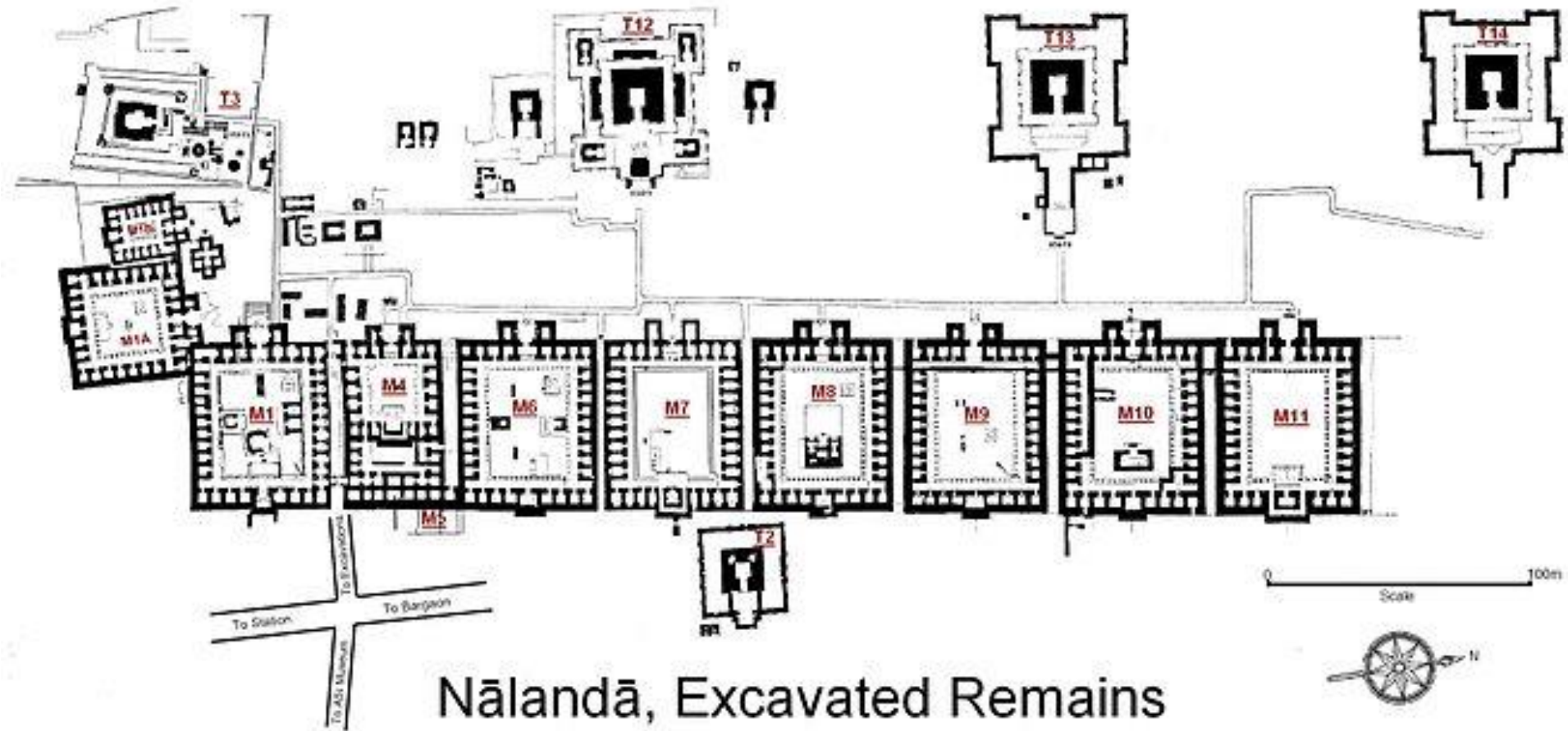
- Historical sources indicate that the University had a long and illustrious life which lasted almost continually for 800 years from the fifth to the twelfth century CE. It was a completely residential university believed to have 2,000 teachers and 10,000 students. The Nalanda ruins reveal through their architectural components the holistic nature of knowledge that was sought and imparted at this University. It suggests a seamless co-existence between nature and man and between living and learning.



Ruin of Nalanda

- The profound knowledge of the Nalanda teachers attracted scholars from places as distant as China, Korea, Japan, Tibet, Mongolia, Turkey, Sri Lanka and South East Asia. These scholars have left records about the ambience, architecture and learning of this unique university. The most detailed accounts have come from Chinese scholars and the best known of these is Xuan Zang who carried back many hundred scriptures which were later translated into Chinese. Close to the end of the twelfth century Nalanda was destroyed by invaders.





Only 10 % of area is found till today and excavation were done by ASI. In June 2016 the site was declared as UNESCO world heritage site.

Vision & Strategies

- There is now a perfect opportunity to recreate the hallowed universalism of Nalanda as a centre of knowledge.
- Asian countries are coming together to forge a continent based on the foundations of peace and harmony. The decision of the East Asian Summit in 2007, at its meeting in Cebu, Philippines, to endorse the plan to re-establish the Nalanda University underscores the commitment to these values. The aim is to rebuild Nalanda as a seat of knowledge.



Proposed Campus

Site Justification

- The site of the new university is around 18 km from the ancient NALANDA MAHAVIHARA.
- The backdrop to the site are the wonderful hills – The Rajgir Hills
- Rajgir hills are not ordinary hills, many pilgrimage sites are located like –
 1. Vishwa Shanti Stupa
 2. Kundalpur – Birth Place of Lord Mahavir
 3. Pawapuri – Holy place for Jain community
- It was at this place Gautam Buddha spent several months preaching : the sacred ground.
- This site will help students and teacher connect with the legacy of the past.



Kundalpur (Birth place of Lord Mahavir)



Actual Site photograph

POTENTIAL

1. Site will provide a platform to the students and faculties to witness the glory of past
2. Nearest railway station is 3 km from the proposed site and a proposed airport is situated 6 km from the site.

CONSTRAINTS

1. It will be difficult to attract top students and faculties to the sub-urban part of Bihar.
2. Presently the site is in the middle of nowhere. It is surrounded by farm fields.

Neighborhood level Significance

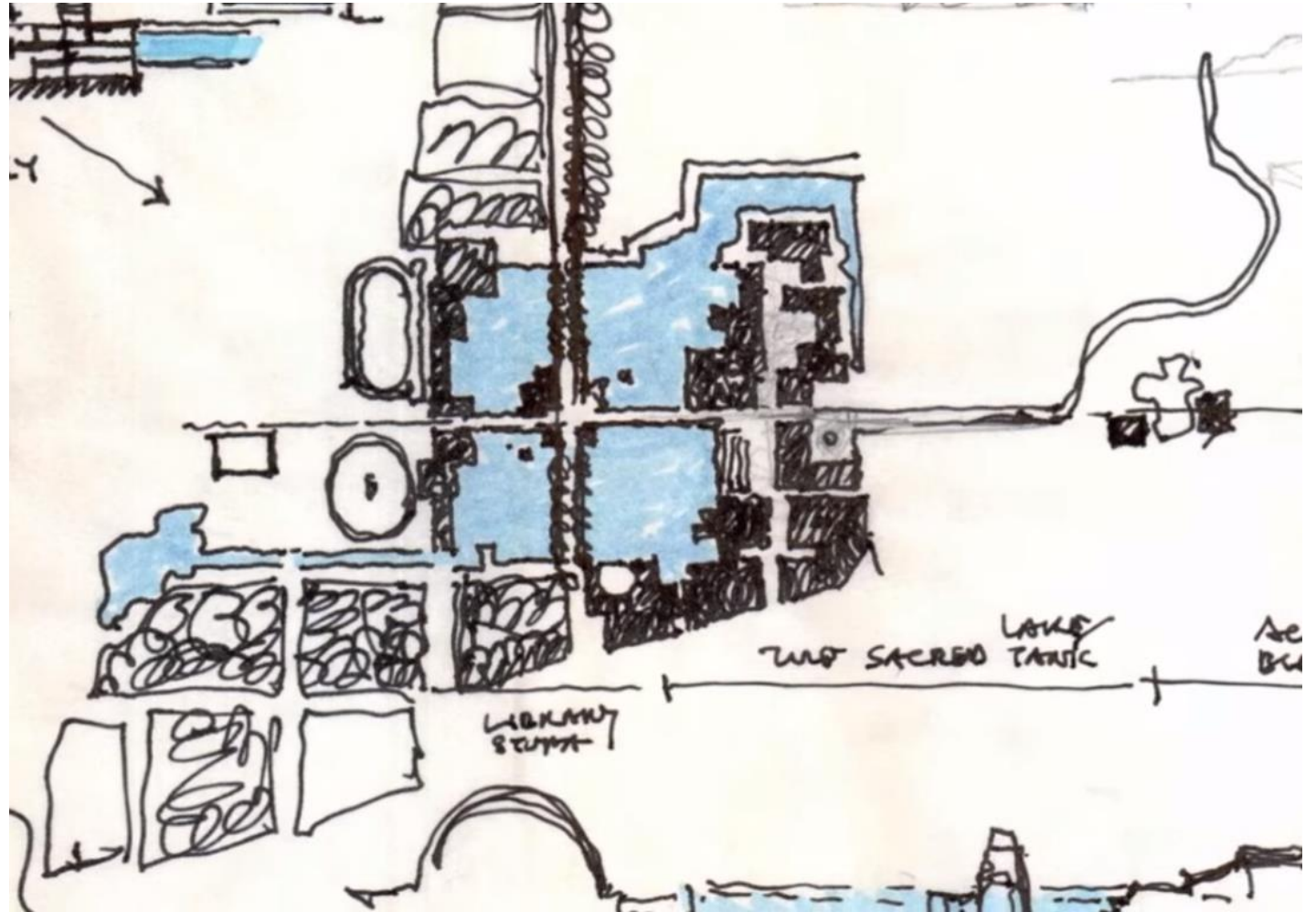
- Site is located around small villages.
- These villages will also get benefit because of university and other activities which are proposed for university.
- They will get job while construction and after construction their land prices will increase.
- Design also respect their existence and facilities also provided for them .



Villages near site

Concept

- Kamal sagar is inspired from ancient Indian temple towns which has water built around tank and harvest rain water.
- The tank is bisected by pathways in all the four cardinal direction .
- At the intersection library is located as heart of the site.
- From the center of the site the concept of kaal chakra came.



Conceptual sketch



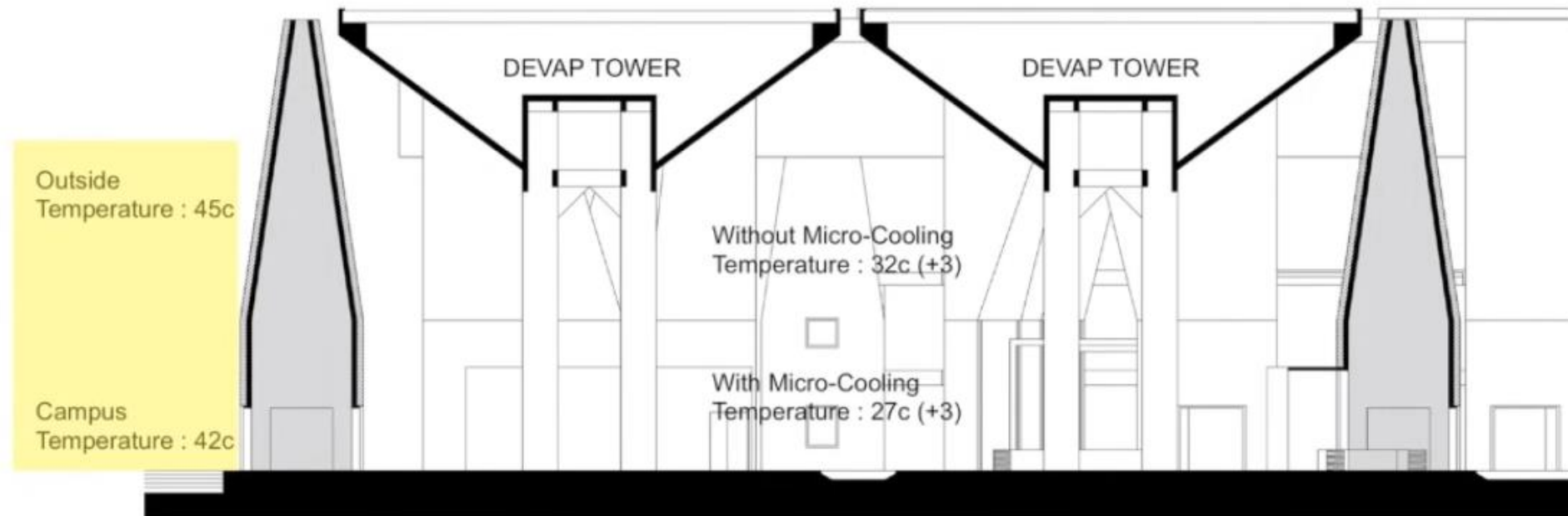
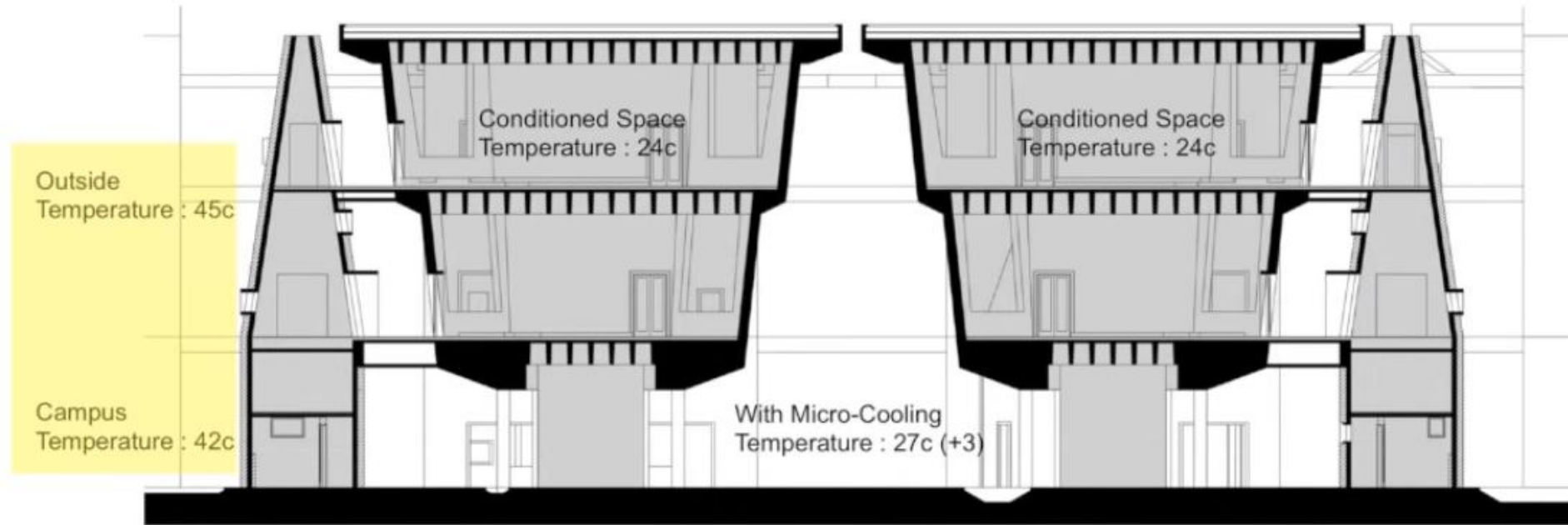
Gopuram

- Gopurams or gateways are also become the source of inspiration for the campus.
- Buffer spaces are inspired from the verandah of traditional Indian architecture



Brick corbel in Nalanda

- The different brick corbel opening of the old Nalanda inspired the new form of campus.
- Bodhi tree also used in campus to create a spiritual feel on site.



Section showing classroom and buffer spaces

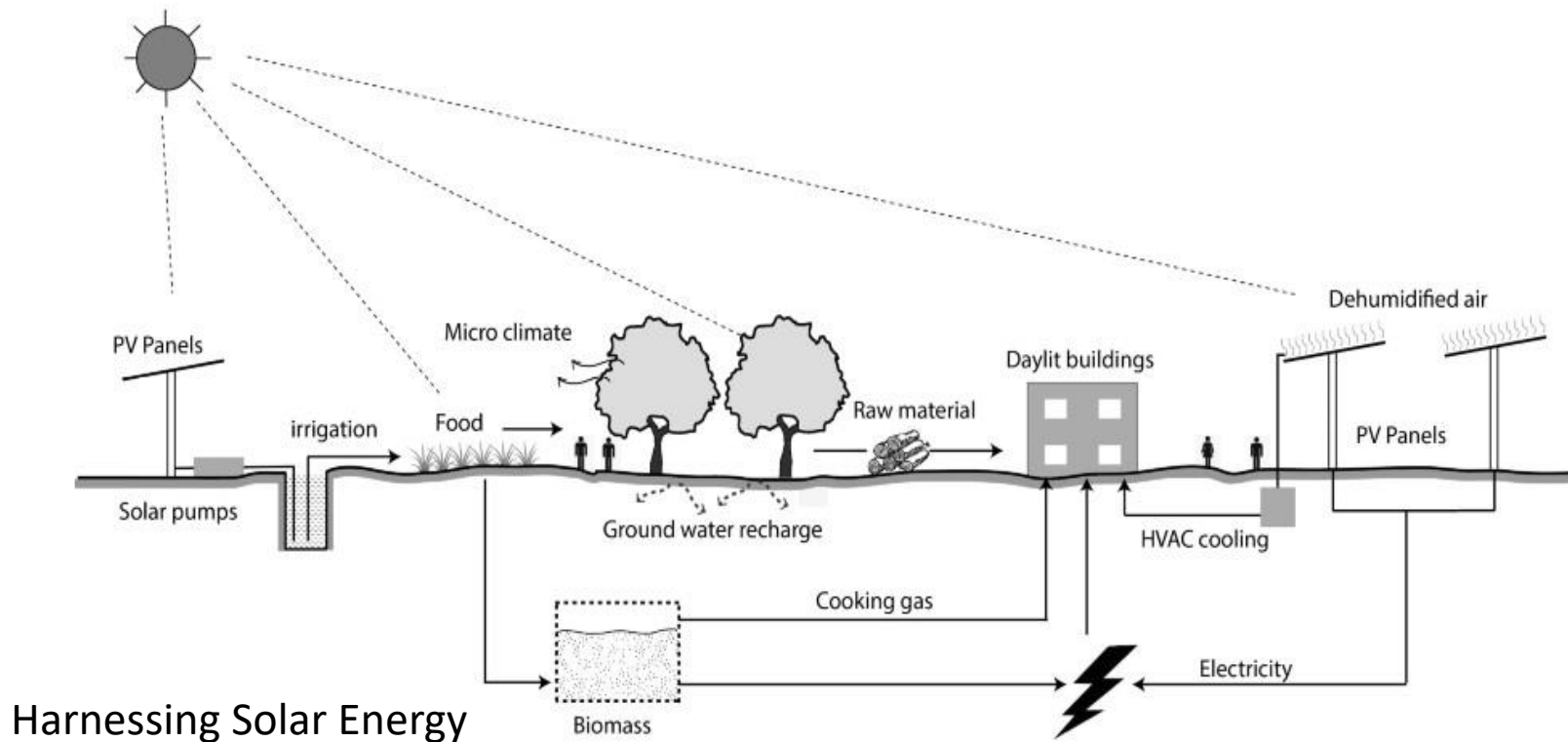
Environmental Impact

- It is the endeavour of the University to develop a sustainable campus which will have a net zero or near zero environmental impact, the first of its kind in the world; special attention was paid to the manner in which each of the schemes addressed the issues of sustainability.
- The master plan integrates sustainable practices, from site planning of the to both reduce consumption of natural resources and minimize dependency on off-site building materials.

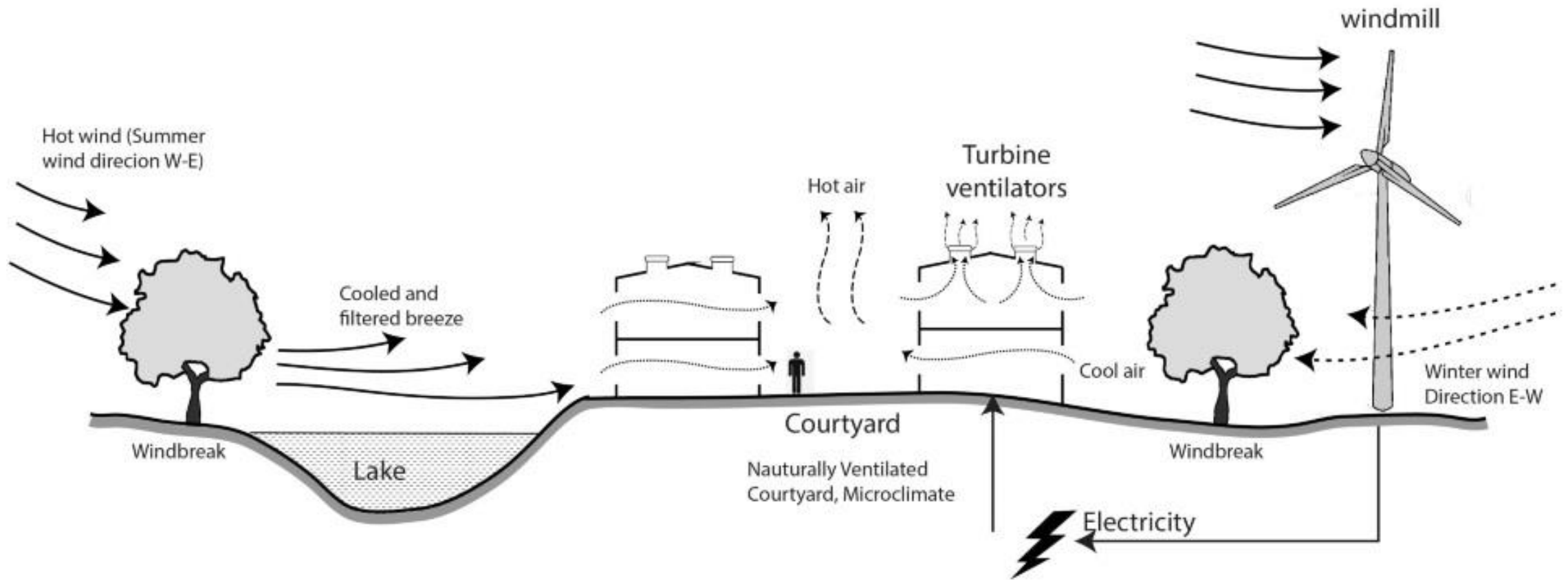


Resource Cycle

- The intent is to create a model campus plan that aims to replicate the spirit of Nalanda that endured in its relevance for 800 years by embracing environmental strategies that are simple, efficient, and appropriate to the place.
- The whole master plan in itself is transitional, as it demonstrates the integration of the campus into a large eco-system of the site.



- By limiting vehicular access to the perimeter of the site, the plan enables 80% of the campus to be reached by walking within 10 minutes



- Wind Energy is also harnessed and taken in consideration while designing the built spaces.
- Lakes are provided in abundance which have a positive impact on micro – climate.

Special Design Features

- Site falls in the cusp of seismic zone 4 where earthquake is very likely to occur.
- Thick walls are provided in the upcoming campus. This concept has been taken from old Nalanda university. Thick walls resist the up thrust of water in sub – soil. Thick walls also resist earthquakes better.
- Sub – soil water level is quite high hence pile foundation is used in the whole campus. In total around 1500 piles have been inserted till dated.
- Compressed Stabilized Earth Blocks (CSEB) are used for superstructure and Burnt Bricks for substructure.

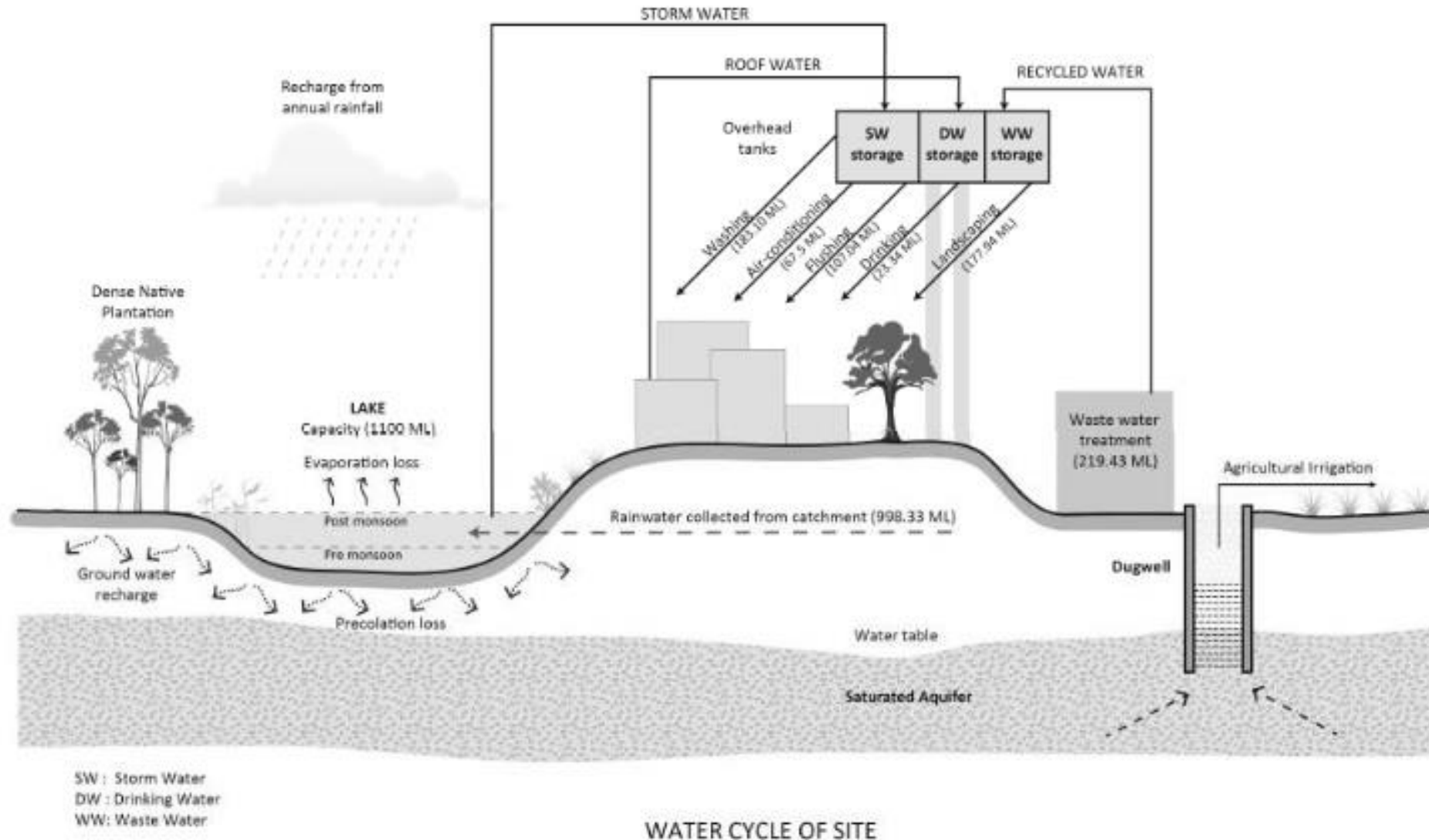


Lotus Pools made after excavation of Water Bodies

Thick walls of old Nalanda university

- Architect has also mentioned quarries used for excavation to make into pools of Lotus.
- It will make them the places of aqua – culture.

- It is also very important to understand the cycle of water – harvesting and how they are connected with buildings.



Process & Procedure

Pre – Construction Process

- Nalanda University launched its Architectural Design Competition in 2012 for the Master Plan of the campus and architectural design of buildings and facilities that will come up in Phase-1 of its development.
- The total of 79 entries received were examined by a Selection Committee which declared 8 of them as pre-qualified for participation in the main competition.
- An international Jury comprising four renowned architects and three members of the Governing Board of the University was constituted to assess the submissions by these eight pre-qualified architects.
- Vastu Shilpa Consultants were selected as the winner of this competition.

Process & Procedure

Construction Process

- While pondering over the question of sacred, Architects initially worked upon the connections that were existing on site.
- There were several villages that surrounded the site. A lattice of embankments were observed from survey drawing that carry the path which linked the villages. These embankments were retained in the master plan.
- This proper planning strategy saved on infrastructure and allowed reduction in capital and operational cost.
- Campus will be developed in three phases : Phase 1, Phase 2 and Phase 3.



- Since the embankments were retained on site, it was compulsory to construct the roads in initial phase. Water Bodies like Ahar and Pyne were also among the first structures on site.
- Reason behind constructing these structures is the need of soil for making of burnt bricks.
- A kiln is also in process of construction for manufacturing of bricks.
- Till date Roads, Ahars, Site Office, Balancing Tanks and Insertion of Piles have been done.





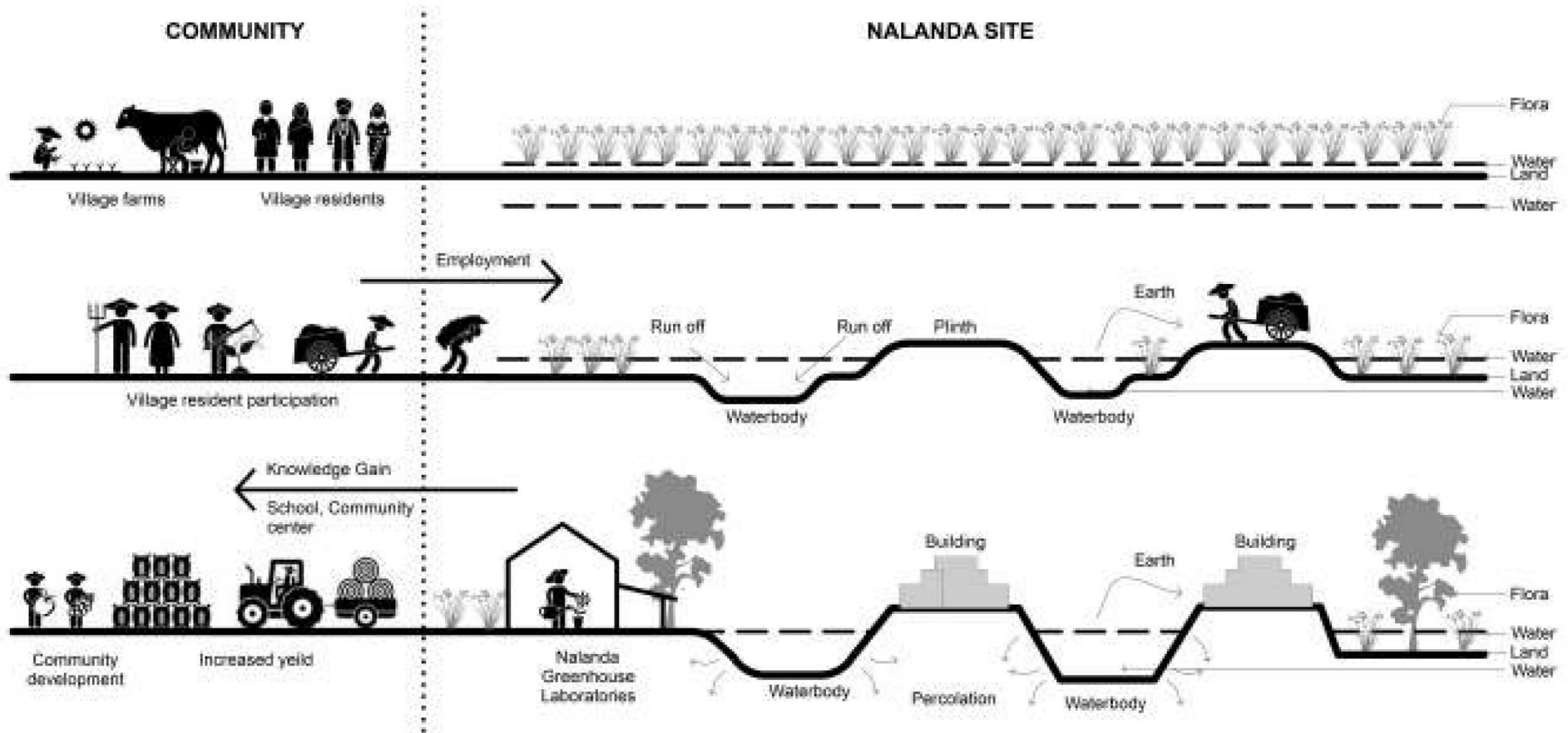
Master plan

Policy Recommendation

- The University is governed by the Nalanda University Act (2010).
- As seen from above slides, it is apparent that the site was originally agricultural farms. Hence, to acquire the land and to fund the project phase- wise, Govt. of India made an act called Nalanda University Act. Following are the notifications printed in the Gazette.
- Re-zoning was done and agricultural land was converted in institutional land. Farmers were re-located from site and land was provided to them according to this act.

1. [Nalanda University Act, 2010](#)
2. [The Gazette of India Notification, November 2012](#)
3. [The Gazette of India Notification, November 2013](#)
4. [The Gazette of India Notification, November 2016](#)

Public Participation



Integrating society as generator and conservator of the campus

Issues and Challenges

- There were many challenges faced during the construction of project. Some of them are :
 1. Improper calculation of soil excavation for manufacturing of burnt bricks.
 2. Survey of water table was not proper so the excavation of balancing tank had to be stopped.
 3. Flooding because of water run-off from hills is a serious problem and to tackle it, Flood Holding Area has been developed. But they will function only when the landscape grows.
 4. Many Asian countries including Singapore, Thailand, Laos had agreed to finance the project but till date only 12 crores have been financed by these countries.
- There are several administrative also which has led the improper functioning of the University.



- Because of miscalculation of ground water – table, excessive excavation was done which resulted in flooding on site.
- It is impossible to pursue construction during monsoon because of heavy rainfall and high water run – off from the neighboring Rajgir hills.

Governance Concern

- The President of India is the Visitor of Nalanda University.
- The Chancellor is the head of the University and is also the Chairperson of the University's Governing Board. The Chancellor presides over Governing Board Meetings and Convocations of the University.
- The Vice Chancellor is the principal academic and executive officer of the University.
- The University's Governing Board is responsible for all the policies and directions of the University and management of its affairs. The Board is also responsible for ensuring that the University achieves its goal of becoming internationally reputed as a center of research and new knowledge.

Impact – Relocation and Tourism

- Some small villages are relocated around 6 kilometer near ordnance factory. compassion is given to the relocated people and land is also given to the relocated community
- Tourist will also attract towards new campus who came to visit old Nalanda university and other historical and religious sites.



Target Groups

- The university offers a vast range of courses in Masters program.
- It cater students from India as well as from other countries also. Special quota has been kept for the students of countries which have agreed to fund the project.
- Youths from various countries will also be welcomed at the university.
- New Nalanda University will be able to accommodate 2500 students and 500 faculties.
- Efforts have been made to attract top faculties and students from all over the world.

Funding of the Project

- Nalanda University, which has set its first academic session in 2014, has bagged a Rs 2,727 crore proposal for financial support over a period of 12 years (2010-11 to 2021-22) approved by The Cabinet Committee on Economic Affairs.
- Out of the Rs 2,727 crore, Rs 1,750 will be used to fulfill the capital cost of the project whereas Rs 977 crore will be diverted towards the University's recurring expenditure for a period of eight years from 2014-15 to 2021-22.
- The main University campus which fosters housing facilities for the teaching and non-teaching staff is being constructed on a 455 acres plot. The plot has been leased out by the Bihar Government, free of cost, for 99 years.

- Adding to the funds received from the Indian government, the University has received voluntary contributions from countries such as China, Thailand and Laos which account to Rs 7.5 crore. Along with monetary help, the University has also received contributions to aid its infrastructural development. Singapore has offered to build a library at a cost of Rs 32.5 crore.
- Japan has guaranteed assistance to renovate the highways leading to the University whereas Australia has committed Australian RS 5 crore for a Chair in the School of Ecology and Environment Studies.
- The University will start chipping in its share towards the recurring expenditure once it starts generating funds after teaching begins. It will also receive endowments through public-private partnerships; an Endowments Committee has been constituted to raise funds through the same.

Thank You